ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

[redacted]

MARKING SCHEME

JUNE 2017

CHEMISTRY 9189/1

1 (a) (i) The ratio of product of the concentration of products (raised to the appropriate powers) to the product of concentration of reactants (raised to the appropriate powers); [1]

eq/m

/AW /(A) equation [redacted] eq/m

Emphasis is on the term "Eq/m"

(ii) Temperature; [1]

$[A]^m[B]$ eq/m

(iii) Equilibrium constant of an exothermic reaction decreases with increase temperature ~~and that of~~ an endothermic reaction increases with increase temperature; AW [1]

(b)

2Y	+	Z	$\rightleftharpoons$	X
1.00		0.75		0.00
at equi				
0.70		0.60		0.15;

[1]

$$K_c = \frac{0.15\ \text{moldm}^{-3}}{(0.70\ \text{moldm}^{-3})^2 \bullet (0.60\ \text{moldm}^{-3})}$$

ecf

working : 1

$= 0.510;\ \text{mol}^{-2}\text{dm}^{6};$

Ans w/ correct units 1

1 mark – ans

1 mark – units [2]

OR $0.510\,V^{2}\,mol^{-2}dm^{6}$

(c) (i) $\cancel{14H^+} + Cr_2O_7^{2-} + 6Fe^{2+} \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O;$ [1]

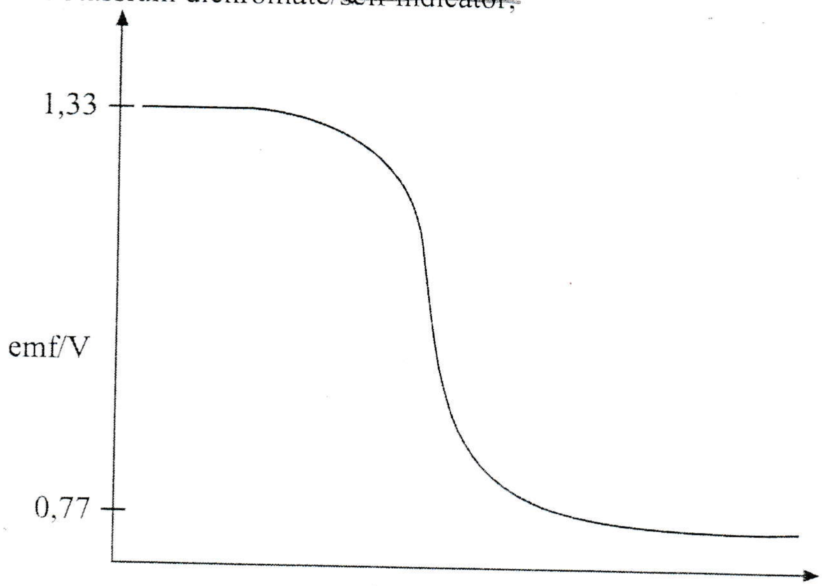
R $H_2O_{(aq)}$

(ii) 1.33 V – 0.77 V

= +0.56 V;

(Ans w working) [1]

(iii) Potassium dichromate/~~self indicator;~~ [1]



[3]

Axis labelled 1
Correct Shape 1
Points 1.33 V and 0.77 V – 1

Curve starting at and ending at l

2 (a) $C_xHy + \left(x + \frac{y}{4}\right)O_2 \rightarrow xCO_2 + \frac{y}{4}H_2O$

10	50 – 5	30
$\frac{10}{10}$	$\frac{45}{10}$	$\frac{30}{10}$
1	4.5	3

(i) volume of oxygen used = 50 – 5

= 45 cm^3; [1]

(ii) carbon dioxide produced = 35 – 5

= 30 cm^3; [1]

(iii) x = 3; [1]

$\frac{y}{4} + 3$ = 4.5 [1]

y = 6; [1] [1]

Formula = C_3H_6; [1] [1]

(b) (i) ΔH_s solution – is the energy change when 1 mole of an ionic substance [1] completely dissolve in water to form a solution of infinite dilution (large amt of H_2O) under standard condition; (298 k, 1 atm, Ignore 1M) [1] [2]

(ii)

$x^+y^-_{(s)} \xrightarrow{\Delta H \text{ soln}} X^+_{aq} + Y^-_{(aq)}$

ΔH latt: $x^+y^-_{(s)} \rightarrow X^+_{(g)} + y^-_{(g)}$

ΔH hydration: $X^+_{(g)} + y^-_{(g)} \rightarrow X^+_{aq} + Y^-_{(aq)}$

with state symbols

1 mark

OR eqn

[2]

exothermically

If ΔH hydration is greater than ΔH lattice, substance dissolves; [1]

$\Delta H_{soln} = \Delta H_{hyd} - \Delta H_{latt}$ 1 mark

ΔH_{hyd} is more exothermic than ΔH_{latt}. 1 mark

(iii) Strong forces of attractions due to high charge densities in both Mg^{2+} and O^{2-}; / ΔHlatt more exothermic than ΔHhyd. [1]

MgO forms insoluble $Mg(OH)_2$ which forms a coating preventing further reaction of MgO with water. [1]

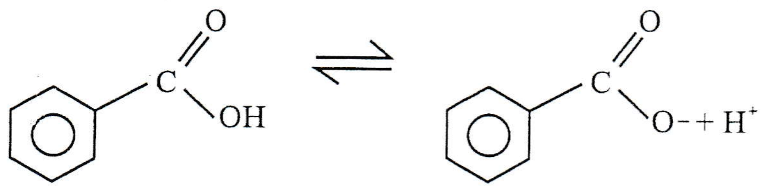
[3 max 2]

3 (a) (i) Solution which resist pH change on addition of small amounts of acid or base; [1]

When little acid $HCO_3^- + H^+ \rightarrow H_2CO_3$; (R) ⇌ and wrong state symbols [1]

When little base $HCO_3^- + OH^- \rightarrow CO_3^{2-} + H_2O$; [1]

(b) (i)



$[H^+]^2 = 6.4 \times 10^{-5} \times 0.02$; [1]

$[H^+] = \sqrt{1.28 \times 10^{-6}}$ M = 1.13×10^{-3} mol dm^{-3}

(ii) pH $= -\log[1.13 \times 10^{-3}]$

(A) 1, 2, & 3 dps. (dps?) $= 2.95$; [1]

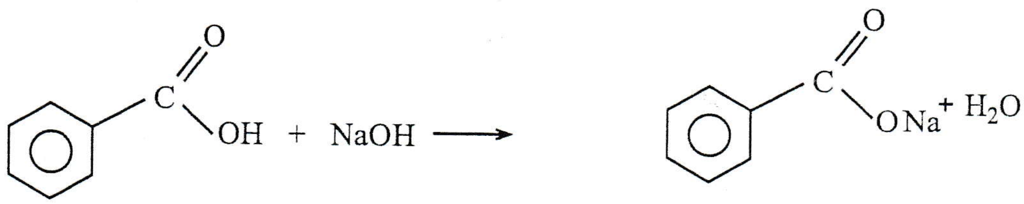
n(sodium benzoate) = $\frac{7.2}{144}$ = 0.05 1 mark

(iii) pH $= -\log 6.4 \times 10^{-5} + \log\frac{[0.05]}{(0.02)}$; [1]

pH $= 4.19 + 0.398$

$= 4.59$; [1]

(iv)



0.02	0.05
0.001	+0.00
0.019	0.051;

$$pH = -\lg 6.4 \times 10^{-5} + \log\frac{(0.051)}{(0.019)}$$

$= 4.19 + 0.429$

$= 4.62;$ [1]

(iv) pH change $= 4.62 - 4.59$ e.c.f.

$= 0.03;$ [1]

(c) Kw at 25°C $= 10^{-14}$

Kw $= [OH] \bullet [H^+]$

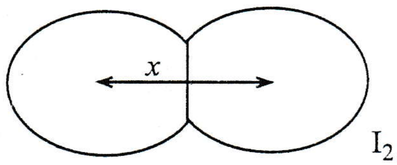
$10^{-14} = [H]^2$ [1]

$[H^+] = 10^{-7};$

pH $= -\lg 10^{-7}$ [1]

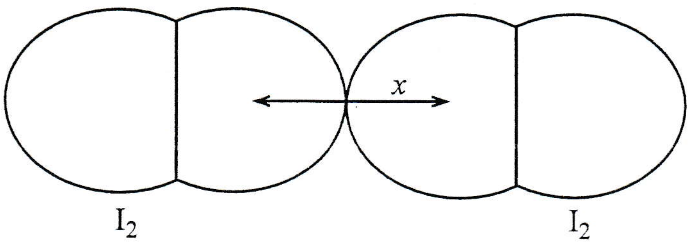
$= 7;$

4 (a) Covalent radius $= \frac{x}{2}$



[1]

Half the distance between two covalently bonded atoms [1]



Half the distance between two iodine atoms not chemically bonded; [1]

• Covalent radius is shorter than VDW radius [1]

(b) $MgCl_2$ is a giant ionic compound with strong electrostatic forces of attraction; [1]

Al_2Cl_6 and $SiCl_4$ are both simple molecular with weak Van der Waal forces; [1]

Al_2Cl_6 has stronger Van der Waal's forces than $SiCl_4$ due to larger dimmer molecules.

(c) Carbon, silicon and Germanium have giant structures; [1]
but down the group atomic size is increasing; [1]
and bond length is also increasing weakening the bond; [1]
This causes decrease in melting point from carbon to Germanium
Tin and lead are metallic; (covalent bonds stronger than metallic) [1]
stronger than Sn ~~Metallic character increases from tin to lead;~~ lead has a closed packing
Thus m.p also increases [1]

5 (a) As refractory materials in the lining of furnaces; Aw [1]
have high melting point; [1]

/used in powerline insulations; [1]
Good thermal and electrical insulators; [1]
/manufacture of glass tiles, dinner ware;
strong, brittle and resistant to attack by chemicals;
/making glasses for solar panels;
(~~resistant to heat + chemical attack;~~) Glass allows sunlight to pass.
(any two)

(b) (i) (discrete)
Contrast CO$_2$ simple molecule with weak Van der Waal's forces; [1]

Compare SiO$_2$ has a giant structure with strong covalent bonds; [1]
~~both~~ intra both are covalently bonded. [1]

(ii) Both CO_2 and SiO_2 are acidic; thus react with bases [1]

$$CO_{2(g)} + 2OH^-_{(aq)} \rightarrow CO_3^{2-} + H_2O_{(l)}$$ [1]

$$SiO_{2(g)} + 2OH^-_{(aq)} \rightarrow SiO_3^{2-} + H_2O_{(l)};$$ [1]

(c) +4 is more stable than the +2 oxidation state; inert pair effect ~~less~~ ~~strong~~ weak. [1]
due to inert pair effect which becomes stronger down the group; [1]

6 (a)

C	O	H	
$\frac{62.1}{12}$	$\frac{27.6}{16}$	$\frac{10.3}{1}$	
$\frac{5.175}{1.725}$	$\frac{1.725}{1.725}$	$\frac{10.3}{1.725}$	[1]
3	1	6	

Empirical formula C_3H_6O; [1]

$n(NaOH) = 8.6 \times 10^{-3} \times 0.1$

$= 8.6 \times 10^{-4}$

$n(Acid) = n(NaOH) = 8.6 \times 10^{-4}$; [1]

$M_r = \frac{0.1}{8.6 \times 10^{-4}}$

$= 116$; [1]

AW

$(C_3H_6O)n = 116$

$n = \frac{116}{58} = 2$

Molecular formula = $C_6H_{12}O_2$; [1]

(b) (i) Stage 1 Reflux; ethanol ethanoic KCN; [2]

Stage 2 Reflux; dil H_2SO_4; [2]

(ii) Hydrolysis; /oxidation [1]

(iii) Q - $CH_3(CH_2)_4CN$; [1]

P - $CH_3(CH_2)_4COOH$; [1]

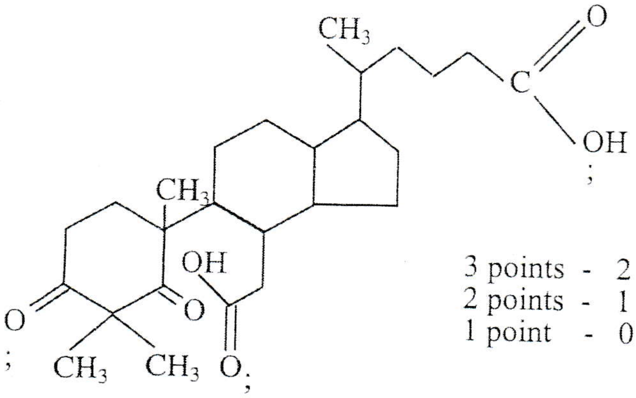
7 (a) 5 ~~8~~ chiral centres; [1]

(b) (i) Alkene; [1]
hydroxyl; [1]

(ii) add Bromine in the dark [1]
Bromine decolourised (alkene); [1]

add PCl_5 - [1]
white fumes (alcohol); [1]
(Na/K - effervesce) 1

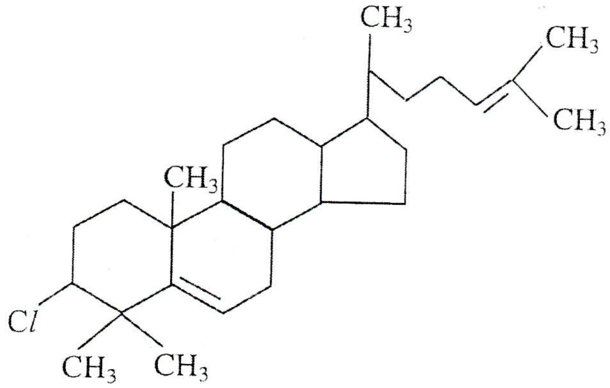
(c) (i)



3 points - 2
2 points - 1
1 point - 0

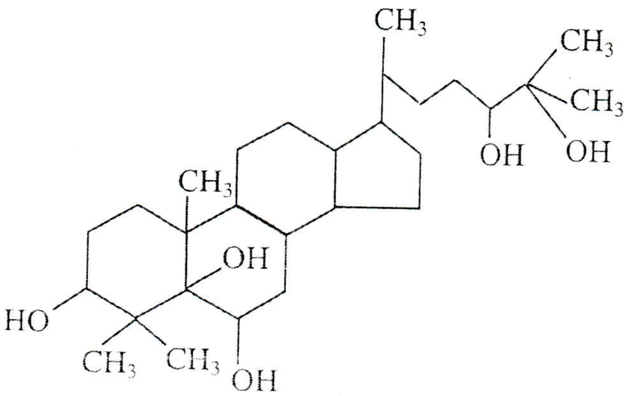
CH_3
$C=O$
CH_3
[1]

(ii)



[1]

(iii)

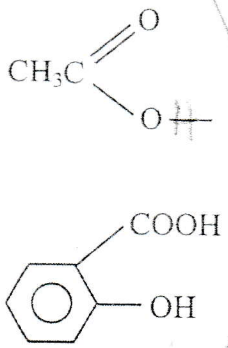


[1]

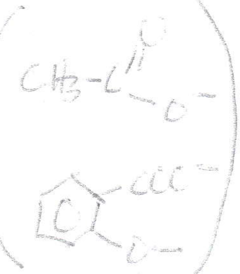
8 (a) ~~anode;~~ Amide [1]
ester; [1]

(b) (i)

in acid

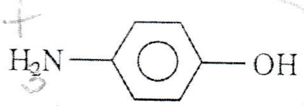


in base



[1]

link str to conditions given



CH_3COOH

in acid

CH_3COO^-

in base. [1]

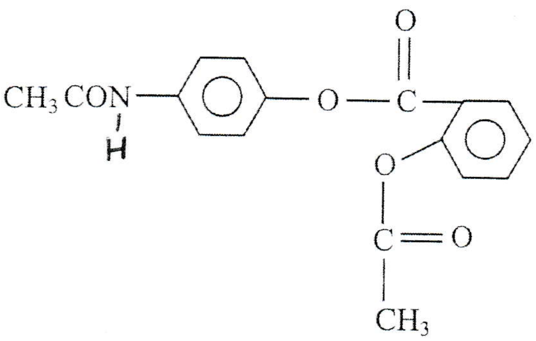
(ii) Reagents Conditions

dil HCl/NaOH; boil/reflux; [2]

(c) Add Na_2CO_3 to the two samples; [1]
Paracetamol – no observable change in Aspirin – effervescence and gas which
Turns limewater milk produced; [1]
(Br_2/e) decolourises but aspirin does not due to ...

(d) Both drugs slightly soluble in water; due to larger hydrophobic groups; [1]
Paracetamol

(e)



[1]